

SATQUERY-AI MISSION REPORT

GEOSPATIAL BI-TEMPORAL CHANGE DETECTION & EVIDENCE REPORT • SMART INDIA HACKATHON

Mission ID:	4c7ada59-25f7-44c9-ad04-53c7fe8a4c4d	Generated:	2026-09-08T21:15:21
Mode:	bi_temporal	Detector Threshold:	0.70 (LEVIR-CD Calibrated)
Checkpoint:	best_model.pt (SiameseUNet)	Confidence:	N/A — Uncalibrated

1. EXECUTIVE SUMMARY

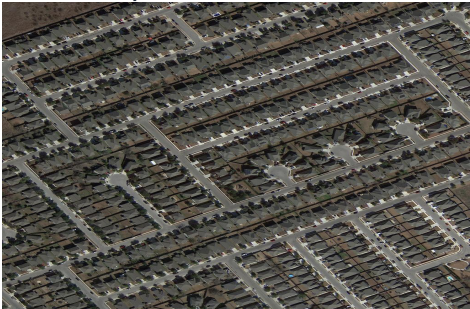
SatQuery-AI completed bi-temporal change analysis for mission ID 4c7ada59-25f7-44c9-ad04-53c7fe8a4c4d. The Siamese U-Net detector identified 26.61% changed pixels (279,050 of 1,048,576 total pixels) between the supplied acquisitions.

2. VISUAL EVIDENCE & SENSOR APERTURE

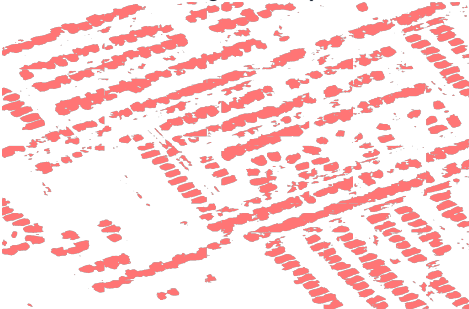
T1: Before Acquisition



T2: After Acquisition



Siamese U-Net Change Heatmap



3. QUANTITATIVE CHANGE DETECTION & SPATIAL ANALYTICS

Total Scene Pixels	1,048,576	Changed Pixels	279,050 (26.61%)
Unchanged Pixels	769,526 (73.39%)	Model Threshold	0.70
Coordinate System (CRS)	Not available	Spatial Resolution (GSD)	Not available
Physical Changed Area	Unavailable (No reliable GSD)	Total Physical Scene	Unavailable (No reliable GSD)

4. CONNECTED COMPONENT HOTSPOTS (TOP LOCALIZED CLUSTERS)

Cluster #	Pixel Area	Share of Change	Centroid (px: X, Y)	Bounding Box (px: x1, y1, x2, y2)
Cluster 334	12,160 px	4.36%	(755.9, 675.3)	[0, 0, 0, 0]
Cluster 123	9,441 px	3.38%	(233.7, 303.7)	[0, 0, 0, 0]
Cluster 10	9,014 px	3.23%	(156.4, 88.0)	[0, 0, 0, 0]
Cluster 441	8,069 px	2.89%	(344.1, 883.5)	[0, 0, 0, 0]
Cluster 126	7,308 px	2.62%	(415.3, 293.6)	[0, 0, 0, 0]

5. REGION OF INTEREST (ROI) INVESTIGATION

ROI investigation was not performed.

6. AI VISION-LANGUAGE INTERPRETATION (CHANGE VQA)

Status: Executed via Domain-Adapted VLM	Confidence: N/A — Uncalibrated
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AI Interpretation:

Change Detection Report (Siamese U-Net Model) **Trained SiameseUNet model detected change across 26.6% of the image area** (279,050 of 1,048,576 pixels), indicating high scene change (severity: **high**). The remaining 73.4% of pixels were classified as unchanged. **Model details:** Lightweight Siamese U-Net (~490K parameters) trained on the LEVIR-CD building change detection dataset (1024×1024 RGB image pairs, binary change masks). Inference uses overlapping 256×256 tile sliding with 32px overlap. **Interpretation:** This result reflects the model's learned change representation. The model was trained on building-related changes in bi-temporal satellite imagery. Results outside that domain may be less reliable. *Analysis performed by trained SiameseUNet checkpoint: C:\Users\Lenovo\Desktop\SatQuery-AI\backend\checkpoints\best_model.pt (operating threshold: 0.70; confidence: uncalibrated / null)* ### Bi-Temporal Scene Change Interpretation **Qualitative Visual Interpretation (VLM):** [VLM temporal interpretation insufficient: The model described static scenes independently without explicit temporal transition or change-oriented dynamics. Raw output: "Before: Agricultural fields After: Urban development"] **VLM Interpretation Validation:** `INSUFFICIENT\_TEMPORAL\_REASONING` (Static scene descriptions without temporal transition phrasing) **Quantitative Detection Telemetry:** - **Detected Changed Area:** `26.61%` (Severity: **high**) - **Change Detection Threshold:** `0.70` (Siamese U-Net) - **Detector Model:** SiameseUNet (~490K parameters, LEVIR-CD trained checkpoint) - **Confidence:** Not calibrated for this bi-temporal analysis (confidence = null) - **Inference Provenance:** Vision-Language Model (local:HuggingFaceTB/SmoVLM-500M-Instruct)

Evidence Grounding:

- Model Limitation: Current change detector is validated primarily on LEVIR-CD-style optical building/structural change.
- Inference mode: trained SiameseUNet model checkpoint (C:\Users\Lenovo\Desktop\SatQuery-AI\backend\checkpoints\best_model.pt).
- Tile size: 256×256px with 32px overlap (probability averaging).
- Inference threshold: 0.70 (operating decision boundary on LEVIR-CD validation split; not a model confidence score).
- Changed pixels (model prediction): 279,050 / 1,048,576 (26.61%).
- Unchanged pixels: 769,526 / 1,048,576 (73.39%).
- Severity label: high (heuristic: low <5%, moderate 5–25%, high >25%).
- Reference image dimensions: 1024×1024 px.
- Change mask overlay saved to: /api/results/4c7ada59-25f7-44c9-ad04-53c7fe8a4c4d_changemap.png
- Confidence score is not produced by the change detection model; confidence field preserved as null.
- Output is from a trained model checkpoint, not fabricated or from a template.
- [Change Detector] Model: SiameseUNet (~490K params, LEVIR-CD trained checkpoint).
- [Change Detector] Measured change area: 26.61% (severity: high, threshold: 0.70).
- [Change Visualizer] Reasoning image passed to VLM: 2-panel (T1 | T2) (2048x1024px).
- [Change Visualizer] Evidence visualization synthesized: 3-panel contour strip (3072x1024px).
- [VLM Reasoning] Provider: local (SmoVLM with domain-adapted LoRA).
- [VLM Reasoning] Model 'HuggingFaceTB/SmoVLM-500M-Instruct' ready (34.0s load; params=507.5M, device=cpu).
- [VLM Reasoning] PEFT LoRA domain adaptation active from checkpoint: checkpoints\vqa_lora_experiment_01\best (trained on BigEarthNet).
- [VLM Reasoning] Model does not emit a calibrated confidence score; confidence=null.
- [VLM Reasoning] [validation] Confidence not produced by model; confidence field preserved as null.
- [VLM Validation] Output flagged as INSUFFICIENT_TEMPORAL_REASONING: Lacks temporal transition phrasing (raw: 'Before: Agricultural fields\nAfter: Urban development').
- [Integrity] The change mask was generated by SiameseUNet; qualitative interpretation was processed by domain-adapted VLM.
- Model does not emit a calibrated confidence score; confidence=null.

7. EXECUTION TRACE & MODEL PROVENANCE

Step	Title	Status	Execution Detail
step-1	Query Received	DONE	Query accepted: "What land-use changes occurred between these two dates?" Mode: bi_temporal Images: 2
step-2	Input Validation	DONE	test_45.png: PNG Modality=rgb_optical file_exists=✓ (2582385 B) 1024×1024px bands=3 ; test_45 (2).png: PNG Modality=rgb_optical file_exists=✓ (2219404 B) 1024×1024px bands=3 ; Pair Compatibility: COMPATIBLE ; Temporal: unknown
step-3	Task Classification	DONE	Detected intents: [vqa: 2.00] [change_detection: 2.00] [change_vqa: 2.00] [change_description: 1.40] Tasks: vqa, change_detection, change_vqa, change_description

step-4	Tool Selection	DONE	change_detector[REAL] → Bi-temporal Change Detector 0.3.0-p0 change_vqa[REAL] → Change-VQA Language Model 0.3.0-p0
step-5	Parameters	DONE	change_detector: algorithm=siamese-unet-model, tile_size=256, tile_overlap=32, threshold=0.7 change_vqa: temperature=0.2, max_tokens=768, provider=local
step-6	Processing	DONE	Bi-temporal Change Detector[REAL]: 10381ms Change-VQA Language Model[REAL]: 86579ms real SiameseUNet model change map generated (LEVIR-CD trained checkpoint)
step-7	Aggregation	DONE	Merging tool outputs Concatenating answer sections Confidence: null (no calibrated model-reported score)
step-8	Completion	DONE	Analysis complete Confidence: null Total elapsed: 97.63s tools=[change_detector=real, change_vqa=real]

8. SENSITIVITY, UNCALIBRATED METRICS & LIMITATIONS

- Confidence is not calibrated across detector and vision-language models (strictly reported as N/A — Uncalibrated).
- Change detection operates on pixel-level contrast via Siamese U-Net without assigning semantic land-cover categories.
- Physical area unavailable — reliable spatial resolution (GSD) metadata was not provided in image headers.
- Geospatial coordinates unavailable — analysis spatial coordinates are reported strictly in pixel space.
- ROI investigation was not performed for this report compilation.